

Chapter 1 – Introduction to Ethical Hacking

What You Will Learn in This Lecture

In this lecture, the following important cybersecurity concepts were covered:

- What is Ethical Hacking?
- What is Hacking?
- Types of Encryption
- Symmetric vs Asymmetric Encryption
- Hashing Concept
- CIA Triad (Confidentiality, Integrity, Availability)
- Hash Values & Integrity Verification
- Types of Hackers
- Phases of Ethical Hacking
- Reconnaissance (Information Gathering)
- Legal Aspects & Cyber Laws
- Penetration Testing vs Ethical Hacking
- Importance of Networking & IT Support Knowledge
- Human Psychology in Ethical Hacking

Introduction to Ethical Hacking

◆ What is Hacking?

Definition

Hacking means identifying weaknesses or vulnerabilities in a system, network, application, or even a person, and gaining unauthorized access to perform malicious or unauthorized activities.

These activities may include:

-  Copying data

-  Deleting data
 -  Encrypting and locking files
 -  Asking money to decrypt the files (Ransomware Attack)
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Example of a Ransomware Attack

Suppose a hacker enters a company's system and encrypts all important files.

Now employees cannot open any file because everything is locked.

The attacker then demands money to provide the decryption key.

This is called a:

Ransomware Attack

Real-Life Example

A hospital system gets hacked and all patient records become locked.

The attacker says:

“Pay ₹10 lakh in Bitcoin or your files will never be recovered.”

What is Ethical Hacking?

Definition

Ethical Hacking is the legal and authorized process of testing systems, networks, or applications to find security weaknesses before malicious hackers exploit them.

Ethical hackers help organizations improve security.

Goal of Ethical Hackers

- Find vulnerabilities
- Report them to the company
- Help fix security issues

- Prevent cyber attacks
-

Daily Life Example

Imagine your house has weak door locks.

A trusted security expert checks your house and tells you:

“This window can be easily broken. Install stronger locks.”

That person is similar to an ethical hacker.



Ethical Hacking Requires Knowledge of:



IT Support

You must understand:

- Operating Systems
 - Software Installation
 - File Systems
 - User Permissions
 - Troubleshooting
-



Networking

Networking is extremely important in cybersecurity.

You should know:

- IP Address
- MAC Address
- DNS
- Routers & Switches
- TCP/IP
- Ports & Protocols

- Firewalls

Because hacking mainly happens through networks.



Ethical Hacking is Also About Human Psychology

Cybersecurity is not only technical.

Many attacks target human emotions and behavior.

Hackers study:

- Human mentality
- Emotions
- Trust
- Fear
- Greed
- Curiosity

This is called:



Social Engineering



Example

A hacker sends:

“Congratulations! You won an iPhone. Click here.”

If the person clicks the link, malware may install.

So cybersecurity is also about understanding human behavior.

Encryption

Definition

Encryption is the process of converting readable data (Plain Text) into unreadable format (Cipher Text) to protect it from unauthorized access.

Important Terms

Term	Meaning
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Plain Text	Original readable data
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Cipher Text	Encrypted unreadable data
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Encryption	Locking the data
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Decryption	Unlocking the data
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Key	Secret code used to lock/unlock
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Types of Encryption

Symmetric Encryption

Definition

In symmetric encryption, the same key is used for both encryption and decryption.

Example Scenario

Suppose:

 User A sends "Hello" to  User B.

Before sending:

- The message is converted into encrypted form.
- A secret key locks the data.

Now User B also needs the SAME key to unlock the message.

So:

 **One Key = Encrypt + Decrypt**

This is called:

 **Symmetric Key Encryption**

 **Important Point**

The biggest challenge is:

How to securely share the same key with the other person?

If an attacker gets the key, they can read all messages.

 **Daily Life Example**

A locker where the same key is used to lock and unlock.

 **Popular Symmetric Algorithms**

- AES (Advanced Encryption Standard)
 - DES
 - 3DES
-

2 Asymmetric Encryption

 **Definition**

In asymmetric encryption, two different keys are used:

-  Public Key
 -  Private Key
-

✖ Working

Suppose two machines communicate.

- Receiver has a Public Key
- Sender encrypts data using the Public Key
- Only the corresponding Private Key can decrypt it

📌 Important Point

🔑 Two Different Keys are Used

Key Type	Purpose
Public Key	Encrypt
Private Key	Decrypt

✖ Daily Life Example

A mailbox:

- Anyone can drop letters using the public opening
- Only the owner can open it using the private key

⚡ Popular Asymmetric Algorithms

- RSA
- ECC
- Diffie-Hellman

🔥 Important Clarification About SHA-256

Your lecture mentioned SHA-256 as an encryption algorithm, but technically:

⚠ **SHA-256 is NOT encryption.**

It is a:

Hashing Algorithm

It belongs to the Secure Hash Algorithm (SHA) family.

It is currently one of the most secure and widely used hashing algorithms.

Hashing Concept

Definition

Hashing converts data into a fixed-length unique value called a hash value or digest.

Even a tiny change in the original file completely changes the hash.

VLC Example

Suppose you download:

-  Original VLC from official website
-  Fake premium VLC APK from another website

Even if both files look similar:

- Their hash values will be completely different.

This helps detect tampering or fake files.

Message Integrity Example

Suppose:

 Sender sends file containing “Hello”

Attacker changes it to:

“Bye”

File name remains same.

But the hash value changes completely.

Receiver compares the original hash value.

If hash values do not match:

✗ File has been modified.

This concept is called:

 **Integrity Verification**

Common Hash Algorithms

Algorithm Full Form		Security
MD5	Message Digest 5	Weak
SHA-1	Secure Hash Algorithm 1	Outdated
SHA-256	Secure Hash Algorithm 256	Strong
SHA-512	Secure Hash Algorithm 512	Very Strong

Hash Calculator Tools

You can calculate hash values using:

- MD5File
- VirusTotal
- Pelock Hash Calculator

You can calculate hash of:

- Files
 - Text
 - Passwords
 - Messages
-

⚠ Important Cybersecurity Concept

“Downloading a file does not usually harm the system — executing/installing it does.”

Malware becomes active only when its malicious code runs.

▲ CIA Triad

■ Definition

The CIA Triad is the foundation of cybersecurity.

It contains three major security principles:

Letter Meaning

C Confidentiality

I Integrity

A Availability

Entire cybersecurity works around protecting these three things.

✖ 1. Confidentiality

■ Meaning

Only authorized users should access sensitive data.

It protects privacy and prevents unauthorized disclosure.

✖ Examples

- Your bank details should only be visible to you and your bank.
 - Company confidential documents should not be accessible publicly.
-

Methods to Achieve Confidentiality

- Encryption
 - Access Control
 - MFA (Multi-Factor Authentication)
 - VPN
-

Technologies

- AES Encryption
 - VPN
 - File Permissions
 - Password Protection
-

2. Integrity

Meaning

Data should remain accurate, complete, and unmodified.

No unauthorized changes should happen.

Example

If someone changes exam results or salary data without permission:

 Integrity is lost.

Methods to Maintain Integrity

- Hashing
 - Digital Signatures
 - Checksums
-

✖ 3. Availability

📘 Meaning

Authorized users should always have access to systems and data whenever needed.

✖ Example

If a company website crashes due to DDoS attack:

✖ Availability is lost.

🔒 Methods to Maintain Availability

- Backup Systems
 - Redundant Servers
 - Disaster Recovery
 - Network Security
-

📊 CIA Triad Summary

Principle	Purpose	Maintained By
Confidentiality	Keep data private	Encryption, MFA
Integrity	Keep data accurate	Hashing, Signatures
Availability	Keep systems accessible	Backups, Redundancy



Types of Hackers

☒ 1. Black Hat Hackers



Definition

Malicious hackers who exploit systems for:

- Money
 - Revenge
 - Data Theft
 - Damage
-

☒ Activities

- Ransomware
 - Banking Fraud
 - Data Theft
 - Malware Distribution
-

☐ 2. White Hat Hackers



Definition

Ethical hackers who legally test systems to improve security.

They report vulnerabilities to organizations.



Bug Bounty

Companies reward ethical hackers for finding security flaws.

This reward is called:



Bug Bounty

Example

A white hat hacker discovers a login vulnerability and informs the company before attackers exploit it.

3. Grey Hat Hackers

Definition

Hackers who may access systems without permission but usually do not intend harm.

They often report vulnerabilities later.

Important

Even without malicious intent, unauthorized access is still illegal.

4. Hacktivists

Definition

Hackers motivated by political or social causes.

Example

Environmental activists targeting organizations responsible for illegal tree cutting.

5. State-Sponsored Hackers

Definition

Hackers supported by governments for cyber warfare, espionage, or defense.

Their Work Includes

- Threat Intelligence
 - Cyber Warfare
 - National Security
 - Infrastructure Protection
-

Threat Intelligence (TI)

Threat Intelligence means understanding:

- Who can attack?
 - What type of attack?
 - What vulnerabilities exist?
 - What impact may happen?
-

Phases of Ethical Hacking

Ethical hacking follows a structured process.

Reconnaissance

Definition

Gathering information about the target.

Information Collected

- Name
- DOB
- Mobile Number
- Social Media
- Vehicle Number

- Interests
- Hobbies
- Family Members
- Profession

Hackers use psychology to guess passwords and security questions.

Types of Reconnaissance

Passive Reconnaissance

No direct interaction with target.

Example:

- Social media research
 - Public records
-

Active Reconnaissance

Direct interaction with target.

Example:

- Calling target
 - Sending emails
 - Scanning systems
-

Scanning

Finding:

- Open Ports
- Live Systems
- Running Services
- Vulnerabilities

3 Gaining Access

Exploiting vulnerabilities to enter systems.

Example:

- SQL Injection
 - Password Cracking
-

4 Maintaining Access

Creating persistence using:

- Backdoors
 - Rootkits
-

5 Clearing Tracks

Removing evidence:

- Deleting logs
 - Clearing history
-

6 Reporting

Ethical hackers document:

- Vulnerabilities
 - Risk Level
 - Fixes
 - Recommendations
-

Legal Aspects & Cyber Laws

Important Rule

Unauthorized Hacking = Cyber Crime

Ethical hacking must always be done with written permission.

Important Legal Requirements

- Written Consent
 - NDA (Non-Disclosure Agreement)
 - Respect Privacy
 - Do Not Damage Data
-

IN Indian IT Act 2000

Section 66

Hacking and data theft.

Punishment:

- Up to 3 years jail
 - Or ₹5 lakh fine
-

Section 43

Unauthorized access and data damage.

Section 66F

Cyber Terrorism.

Very severe punishments.

Global Cyber Laws

Law	Country/Region
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CFAA	USA
------	-----

GDPR	European Union
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PCI-DSS	Payment Card Industry
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Penetration Testing vs Ethical Hacking

Criteria	Ethical Hacking	Penetration Testing
Definition	Broad security testing	Simulated attack testing
Objective	Improve overall security	Find exploitable vulnerabilities
Scope	Includes policies & people	Mostly technical systems
Approach	Continuous	Periodic
Example	Organization security audit	Testing website for SQL Injection

Important Note

Penetration Testing is a subset of Ethical Hacking.

Final Understanding of Lecture 1

After this lecture, you should understand:

- ✓ Basics of hacking
- ✓ Encryption & hashing
- ✓ CIA Triad
- ✓ Types of hackers
- ✓ Ethical hacking process
- ✓ Reconnaissance techniques
- ✓ Cyber laws
- ✓ Importance of networking & psychology in cybersecurity

Quick Revision

Topic	Key Point
Ethical Hacking	Legal security testing
Symmetric Encryption	Same key for encrypt/decrypt
Asymmetric Encryption	Public & Private keys
Hashing	Verifies integrity
CIA Triad	Confidentiality, Integrity, Availability
Black Hat	Malicious
White Hat	Ethical
Reconnaissance	Information gathering
Penetration Testing	Simulated attack

Q & A

1. What is Ethical Hacking?

Answer:

Ethical Hacking is the legal and authorized process of identifying vulnerabilities in systems, networks, applications, or organizations before malicious attackers exploit them.

Ethical hackers use the same techniques as attackers, but with proper permission and legal approval. Their main goal is to improve the security posture of an organization.

Responsibilities of Ethical Hackers:

-  Finding vulnerabilities
-  Reporting security flaws
-  Helping fix weaknesses
-  Preventing cyber attacks
-  Creating security reports

Example:

If a company hires a security professional to test whether hackers can break into their website, that process is called Ethical Hacking.

Important Point:

Ethical hacking is always done:

-  With written permission
-  Under legal boundaries
-  For security improvement

Without permission, hacking becomes illegal.

2. What is Hacking?

Answer:

Hacking is the process of identifying weaknesses in systems or networks and exploiting them to gain unauthorized access.

Hackers may perform activities such as:

-  Stealing data
-  Deleting files
-  Encrypting systems (Ransomware)
-  Demanding money
-  Spying on users

Example:

A hacker gains access to a company server and steals customer information.

Interview Tip:

Mention that hacking itself is not always illegal. The legality depends on:

- Permission
- Intent
- Authorization

3. What is Encryption? Explain Its Types.

Answer:

Encryption is the process of converting readable data (Plain Text) into unreadable data (Cipher Text) to protect it from unauthorized access.

Only authorized users with the correct key can decrypt the data.

Types of Encryption

1 Symmetric Encryption

Answer:

In symmetric encryption, the same key is used for both encryption and decryption.

Same Key $\Rightarrow$ Encrypt + Decrypt

Features:

-  Fast
-  Efficient for large data
-  Key sharing is difficult

Common Algorithms:

- AES
- DES
- 3DES

Example:

Wi-Fi passwords often use symmetric encryption.

2 Asymmetric Encryption

Answer:

Asymmetric encryption uses two different keys:

-  Public Key
-  Private Key

Public Key $\Rightarrow$ Encryption Private Key $\Rightarrow$ Decryption

Features:

-  More secure for communication

- 🌐 Used in HTTPS and SSL/TLS
- 🐢 Slower than symmetric encryption

🔧 Common Algorithms:

- RSA
- ECC
- Diffie-Hellman

🌱 Example:

HTTPS websites use asymmetric encryption during secure communication.

🔑 4. Difference Between Symmetric and Asymmetric Encryption 🔒

Feature	Symmetric Encryption	Asymmetric Encryption
🔑 Keys Used	Same Key	Public & Private Keys
⚡ Speed	Faster	Slower
🔒 Security	Less Secure	More Secure
🌐 Usage	File Encryption	Secure Communication
🔧 Examples	AES, DES	RSA, ECC

5. What is Hashing? 🔍

✅ Answer:

Hashing is the process of converting data into a fixed-length value called a hash value or digest.

Hashing is mainly used for:

- 🛡️ Integrity verification
- 🔑 Password storage
- 📁 File verification

📌 Important Point:

Hashing is:

- ❌ NOT encryption
 - ❌ NOT reversible
 - ✅ One-way process
-

🔥 6. What is SHA-256? 🔒

✅ Answer:

SHA-256 is a cryptographic hashing algorithm from the SHA family.

It generates a fixed 256-bit hash value.

📌 Uses:

- Password hashing
- Blockchain
- File integrity
- Digital signatures

⚠️ Important Interview Point:

SHA-256 is NOT an encryption algorithm.

It is a hashing algorithm.

🛡️ 7. What is Integrity in Cybersecurity?

✅ Answer:

Integrity means ensuring that data remains accurate, complete, and unmodified.

If unauthorized changes occur, integrity is compromised.

🔧 Methods to Maintain Integrity:

- Hashing

- Checksums
- Digital Signatures

✖ Example:

If a hacker changes salary data in a company database, integrity is lost.

▲ 8. Explain the CIA Triad 🔒

✅ Answer:

The CIA Triad is the foundation of cybersecurity.

It consists of:

- 🔒 Confidentiality
 - 🛡 Integrity
 - 🌐 Availability
-

🔒 Confidentiality

Means only authorized users can access data.

🔧 Maintained By:

- Encryption
- MFA
- Access Control
- VPN

✖ Example:

Only you should access your bank account details.

🛡 Integrity

Means data should not be modified without authorization.

Maintained By:

- Hashing
 - Checksums
 - Digital Signatures
-

Availability

Means systems and services should remain accessible whenever needed.

Maintained By:

- Backup systems
- Redundant servers
- Disaster recovery

Example:

A DDoS attack affects availability.

9. What are the Different Types of Hackers?

Answer:

Black Hat Hackers

Malicious hackers who perform illegal activities for:

-  Money
-  Damage
-  Data theft

Example:

Ransomware attackers.

☐ **White Hat Hackers**

Authorized ethical hackers who improve security legally.

✚ **Example:**

Bug bounty hunters.

☒ ☐ **Grey Hat Hackers**

Hackers who may access systems without permission but often report vulnerabilities later.

⚠ **Important:**

Still illegal without permission.

🗯 **Hacktivists**

Hackers motivated by political or social causes.

🏛 **State-Sponsored Hackers**

Government-backed hackers involved in:

- Cyber warfare
 - Espionage
 - National security
-

🎯 **10. What is a Ransomware Attack?** 🔒 💰

✅ **Answer:**

Ransomware is a type of malware that encrypts files and demands money for decryption.

✖ Attack Flow:

- 1 Malware infects system
- 2 Files become encrypted
- 3 Victim loses access
- 4 Attacker demands ransom

✖ Example:

A hospital loses access to patient records until ransom is paid.

🛡 Prevention:

- Regular backups
- Antivirus
- User awareness
- Email security

🌐 11. Why is Networking Important in Ethical Hacking?

✅ Answer:

Most cyber attacks occur through networks. Ethical hackers must understand networking to identify vulnerabilities and secure systems.

📁 Important Networking Concepts:

- IP Address
- MAC Address
- DNS
- TCP/IP
- Ports & Protocols
- Firewalls
- Routers & Switches

✖ Example:

Port scanning helps identify running services on a target machine.

12. What is Social Engineering?

Answer:

Social engineering is the psychological manipulation of people to perform actions or reveal confidential information.

Hackers exploit:

- Fear
- Trust
- Curiosity
- Greed

Example:

“Congratulations! You won an iPhone. Click here.”

Common Attacks:

- Phishing
- Fake calls
- Fake emails
- Impersonation

13. What is Reconnaissance?

Answer:

Reconnaissance is the first phase of ethical hacking where attackers gather information about the target.

Information Collected:

- Names
- Emails
- Social media

- Phone numbers
 - Technologies used
 - Employee details
-

Types of Reconnaissance

Passive Reconnaissance

No direct interaction with target.

Examples:

- Google search
 - LinkedIn research
 - Social media analysis
-

Active Reconnaissance

Direct interaction with target.

Examples:

- Port scanning
 - Sending emails
 - Calling employees
-

14. Explain the Phases of Ethical Hacking

Answer:

Phase	Description
 Reconnaissance	Information gathering
 Scanning	Finding vulnerabilities
 Gaining Access	Exploiting vulnerabilities

Phase	Description
 Maintaining Access Persistence/backdoors	
 Clearing Tracks	Removing evidence
 Reporting	Documentation and fixes

Interview Tip:

Always mention that ethical hackers stop after reporting and fixing vulnerabilities.

15. Difference Between Ethical Hacking and Penetration Testing

Ethical Hacking	Penetration Testing
Broad security assessment	Simulated attack
Includes people & policies	Mostly technical
Continuous process	Periodic testing
Security improvement	Exploit verification

Important Point:

Penetration Testing is a subset of Ethical Hacking.

16. What are the Legal Requirements for Ethical Hacking?

Answer:

Ethical hacking must always follow legal procedures.

Important Requirements:

-  Written permission
-  NDA agreement
-  Respect privacy

-  No data damage

 Important:

Unauthorized hacking is a cyber crime.

IN 17. What is Section 66 of the Indian IT Act 2000?

 Answer:

Section 66 deals with hacking and data theft.

 Punishment:

- Up to 3 years imprisonment
 - Or fine up to ₹5 lakh
 - Or both
-

 18. Why is Human Psychology Important in Cybersecurity?

 Answer:

Many cyber attacks target humans instead of systems.

Humans are often the weakest security link.

Hackers exploit:

- Trust
- Fear
- Urgency
- Curiosity

 Example:

Fake banking calls asking for OTP.

 Interview Tip:

Mention that cybersecurity awareness training is extremely important.

🔥 19. What is Bug Bounty? 💰

✅ Answer:

Bug bounty is a reward program where companies pay ethical hackers for responsibly reporting vulnerabilities.

🧩 Examples:

Big companies like:

- Google
- Meta
- Microsoft

run bug bounty programs.

🎯 Benefits:

- Improves security
- Encourages responsible disclosure

🔒 20. What is the Difference Between Encryption and Hashing?

Encryption	Hashing
Reversible	One-way
Used for confidentiality	Used for integrity
Requires keys	No decryption
Plain text can be recovered	Original data cannot be recovered

🧩 Example:

- Encryption → Secure communication
 - Hashing → Password storage
-

Final Interview Tips for Freshers

What Interviewers Expect:

- Basic cybersecurity concepts
 - Networking fundamentals
 - Understanding of encryption & hashing
 - Knowledge of CIA Triad
 - Awareness of cyber laws
 - Communication skills
 - Problem-solving mindset
-

Bonus HR Question

 “Why do you want to enter cybersecurity?”

Sample Answer:

“I am interested in cybersecurity because technology and security are becoming very important in today’s digital world. I enjoy understanding how attacks happen and how systems can be protected. Ethical hacking allows me to continuously learn new technologies, solve security problems, and help organizations stay secure from cyber threats.”
